

Korea's Battery Empire

LG, Samsung SDI, SK On and the Global EV Supply Chain War

Blue Lotus Research Intelligence | August 2026

Classification: Open Intelligence

Executive Summary

Korea's battery triumvirate — LG Energy Solution, Samsung SDI, and SK On — entered 2026 facing the most turbulent competitive environment since the industry's founding. China's CATL and BYD have aggressively expanded global market share, eroding Korea's once-dominant position in European and North American EV supply chains. The Korean Big Three collectively held 13.8% global EV battery market share in January-May 2026, compared to CATL's 37.9% and BYD's 15.8%. Samsung SDI fell out of the global top 10 battery suppliers in Q1 2026.

Yet the situation is more nuanced than the market share headline suggests. Korea's battery companies occupy differentiated strategic positions: LG Energy Solution is CATL's primary global challenger in Western markets; SK On is the exclusive supplier to Ford's critical F-150 Lightning and Volkswagen's North America ID.4; Samsung SDI's solid-state battery pipeline represents the technology bet most likely to reshape the competitive hierarchy before 2030. The \$400B+ battery manufacturing investment in North America — driven by the US Inflation Reduction Act's domestic content requirements — has created a geographically protected market in which Korean suppliers hold a structural advantage over Chinese competitors.

Part I — The Market Landscape: China's Commanding Lead

CATL: The Predator Setting the Pace

Contemporary Amperex Technology Co. Limited entered 2026 with 37.9% of the global EV battery market — a position that would be considered monopolistic in most other industries. CATL's dominance rests on three pillars: massive manufacturing scale (approximately 400 GWh annual production capacity by end-2025), continuous price reduction through the battery learning curve, and proprietary technology (LFP cell chemistry, Kirin battery pack design, sodium-ion chemistry for entry-level vehicles).

CATL's "price war" strategy in 2024-2025 — reducing LFP cell prices to approximately \$55/kWh — forced margin compression across the entire industry. Korean producers, whose costs structure includes higher South Korean labour rates, imported lithium (no domestic lithium reserves in Korea), and proprietary NMC chemistry inputs (cobalt, manganese), cannot match CATL's structural cost position. The Korean response has been to differentiate upward — targeting premium and performance segments where energy density, charging speed, and safety records justify higher ASP.

BYD's Vertical Integration Threat

BYD's 15.8% market share, achieved through complete vertical integration (mining, chemistry, cells, battery packs, and complete vehicles), represents a different competitive model entirely. BYD produces

its own lithium iron phosphate and recently launched its own lithium mining and processing operations in the Atacama Desert. When a BYD vehicle competes with a Kia or Hyundai in the European market, the battery cost advantage that BYD carries through its integration structure is estimated at \$80-120/kWh system-level versus what Korean OEMs pay their Korean battery suppliers.

Part II — LG Energy Solution: The Western Market Champion

Financials and Scale

LG Energy Solution reported 2025 revenue of KRW 30.1 trillion (\$22.3B) — down approximately 18% from 2024's KRW 36.6 trillion, reflecting EV demand softness in North America and Europe and the IRA-driven geographic reorientation of its customer mix. However, H1 2026 showed recovery momentum with quarterly revenue stabilising around KRW 7.5 trillion, supported by IRA Advanced Manufacturing Production Credits (AMPC) of approximately KRW 800-900 billion per quarter.

LGES's manufacturing footprint in North America is the company's most important strategic asset. Joint ventures with GM (Ultium Cells, four plants totaling ~140 GWh planned capacity), Honda (Ohio, ~40 GWh), Hyundai (Georgia, ~35 GWh), and Stellantis (Canada and US) represent a combined investment exceeding \$20B. The IRA's 10% AMPC on battery cells produced in North America provides LGES approximately \$100-130 per kWh in federal credits, creating a structural cost advantage versus any non-North-America manufacturer selling into the US market. This geographic moat is LGES's primary competitive shield against CATL.

Technology Positioning: Cylindrical's Comeback

LGES's strategic bet on large-format cylindrical cells (4680 and 2170 form factors) has been validated by Tesla's adoption and is now gaining traction with additional customers. The cylindrical format offers superior energy density in volumetrically constrained applications (premium EVs, e-VTOL aircraft) and lower pack complexity versus prismatic alternatives. LGES's Ochang factory in Korea produces 2170 cells for Tesla; the Arizona Greenfield (JV with Honda) will produce 2170 for Honda's North American EVs.

The company's timeline to LFP production is another signal. LGES has announced LFP pouch cell production at its Michigan plant from 2025, targeting the mass-market commercial vehicle and affordable EV segment that CATL has dominated. This LFP expansion, delayed several years by strategic hesitation, is essential for LGES to compete below the premium tier.

ESS Pivot: The Battery Empire's New Frontier

LGES's most strategically significant evolution in 2025-2026 is its aggressive pivot to Energy Storage Systems (ESS). With EV growth moderating in Europe and North America, ESS demand — driven by US grid investment, European battery storage mandates, and AI data centre backup power requirements — has become the fastest-growing large format battery application. LGES secured ESS supply agreements with multiple US utilities in 2025, and its Powerplus large-format LFP battery system (for grid applications) has been certified by major US grid operators.

The economics of ESS are favourable for Korean producers: ESS cells require lower energy density (minimising the Korea vs CATL cost-per-kWh disadvantage) while prioritising cycle life and safety characteristics where Korean producers maintain advantages. ESS also faces fewer IRA-driven market distortions — the geographic moat that protects LGES EV cells in North America is less critical in global ESS markets where logistics and financing terms matter more than manufacturing location.

Part III — Samsung SDI: Technology Bet on the Future

The Crisis Quarter: Falling Out of the Top 10

Samsung SDI's fall from the global top 10 EV battery suppliers in Q1 2026 is a watershed moment. The company's global EV battery installation share fell to 3.4% in January-May 2026, from 5.2% a year earlier — a market share loss of approximately 1.8 percentage points in 18 months.

The drivers are structural. Samsung SDI's primary EV battery customer is Stellantis (Jeep, Ram, Chrysler brands), which cut EV production targets in 2025 amid slowing North American EV adoption. SDI's other major customers — BMW (for the iX and i4 series in Europe) and Rivian (the Amazon delivery van program) — were insufficient volume anchors to replace the Stellantis shortfall. Samsung SDI's product mix, skewed toward high-nickel NMC (for range performance), meant it was strategically positioned in the premium segment where demand softened most acutely.

Solid-State: The Transformation Play

Samsung SDI's most important strategic asset is not visible in current market share data. The company's solid-state battery R&D programme — the most advanced among the Korean Big Three — targets commercial production of all-solid-state batteries (ASSB) using Samsung's proprietary oxide electrolyte by 2027, with volume production by 2030. Samsung SDI has announced a dedicated ASSB pilot line at its Cheonan facility, with 50 Wh/kg prototype cells demonstrated to OEM partners in Q3 2025.

Solid-state batteries offer theoretical advantages that would reshape the competitive landscape: energy density of 500+ Wh/kg (versus 300-350 Wh/kg for current NMC), elimination of the flammability risk that plagues liquid-electrolyte lithium-ion, and faster charging times. The challenge — universally acknowledged by Toyota, QuantumScape, Solid Power, and all other ASSB developers — is manufacturing at scale. The production of thin, uniform solid electrolyte layers and the assembly of all-solid-state cells without the liquid electrolyte injection step that current Li-ion manufacturing relies on requires entirely new manufacturing equipment.

If Samsung SDI achieves volume ASSB production before CATL or its Korean competitors, the market position recapture potential is enormous. Toyota, BMW, Volkswagen, and Mercedes-Benz have all signaled willingness to pay substantial premiums for ASSB cells due to safety, density, and charging benefits. Samsung SDI's partnership with Stellantis on a dedicated US ASSB manufacturing facility positions it for the US market specifically.

Part IV — SK On: The Deepest Hole, the Clearest Path

From Loss Leader to Potential Champion

SK On, spun off from SK Innovation in October 2021, has accumulated cumulative operating losses exceeding KRW 3 trillion (\$2.2B) through H1 2026 — a record of sustained loss-making that has severely tested SK Group's commitment to the battery business. The operating losses reflect the company's aggressive greenfield manufacturing buildout: SK On committed to building 18 factories globally between 2021 and 2025, with North American JVs (BlueOval SK with Ford, two plants in Kentucky and Tennessee) representing the primary capital deployment.

The financial picture began improving in Q2 2026. SK On reported its first quarterly operating profit since its spinoff: KRW 48 billion (\$35M) — modest, but psychologically significant. The profitability inflection was driven by IRA AMPC credits from its Kentucky BlueOval SK plant (now at 50% of full 43 GWh capacity) and a recovery in Ford F-150 Lightning production after Ford's 2024 EV production cutbacks.

Ford Dependency and Diversification

SK On's strategic vulnerability is customer concentration. Ford accounts for approximately 55-60% of SK On's revenue — a dependency that amplifies SK On's exposure to Ford's EV programme fluctuations.

Ford's repeated revisions to EV production targets (cutting the F-150 Lightning production target from 150,000 to 100,000 units per year in 2025) directly impacted SK On's BlueOval SK utilisation rates.

SK On's diversification effort targets Volkswagen's US manufacturing programme and the emerging robotaxi/commercial EV segment. Hyundai Motor Group (SK's domestic Korean partner in multiple conglomerates) represents a natural anchor customer for SK On's Korean domestic production. SK On's NMC 9.1 chemistry — achieving 3% more energy density versus its NMC 811 predecessor — positions it favourably for the high-range premium segment where VW and Hyundai compete.

Part V — The IRA Lifeline: North America as Fortress

Manufacturing for Market Access

The US Inflation Reduction Act's battery supply chain provisions have transformed the strategic calculus for all three Korean producers. To qualify for the full \$7,500 EV consumer tax credit, battery cells must be assembled in North America, with increasing proportions of critical minerals from North America or FTA-partner nations. This creates a structural exclusion for CATL and BYD from the US retail EV market — at least through the current IRA provisions.

The Korean response has been massive. LG Energy Solution, Samsung SDI, and SK On together have committed approximately \$50B+ in North American manufacturing investment since 2022. Combined with the AMPC credits (approximately \$35/kWh for cells, \$10/kWh for modules), this investment is financially defensible even at current EV adoption rates. If US EV adoption accelerates toward the 40-50% penetration rates that Biden-era policy targeted, the Korean battery JVs are the primary beneficiary.

FEOC Rules: The Chinese Exclusion Mechanism

The IRA's Foreign Entity of Concern (FEOC) provisions, which exclude vehicles using battery components from Chinese companies (CATL, BYD, CALT, etc.) from the tax credit beginning in 2024, are the single most important policy development for Korean battery competitiveness in the US. CATL's response — building a licensed manufacturing operation with Ford in Michigan (the BlueOval Battery Park) — initially seemed to circumvent FEOC provisions, but subsequent US regulatory guidance in 2024 clarified that licensing arrangements do not resolve FEOC concerns, effectively blocking CATL-Ford from the full credit.

For LG Energy Solution, Samsung SDI, and SK On, FEOC rules are a multi-year competitive gift. The exclusion of CATL from the IRA subsidy framework effectively caps CATL's US market share at OEMs that are willing to forgo the full tax credit — a market that is, by revealed preference, significantly smaller than the credit-eligible segment.

Part VI — Critical Minerals: The Upstream War

Lithium: No Korean Mines

Korea's battery producers have no domestic lithium resources — every gram of lithium used in Korean-manufactured batteries must be imported. This creates structural cost exposure to lithium price cycles and geopolitical disruption. Lithium carbonate prices collapsed from a peak of approximately \$80,000/metric tonne in late 2022 to approximately \$10,000-12,000/metric tonne in H1 2026 — a level approaching or below the marginal cost of production for many lepidolite and clay deposits.

The current low lithium price is beneficial for Korean producers' near-term margins but obscures the long-term supply security question. CATL and BYD have both made significant direct investments in Chilean, Australian, and African lithium mining, securing preferential offtake at below-market prices.

Korean battery companies have been slower to develop upstream mining positions, relying instead on long-term supply contracts with mining companies. POSCO Holdings (through POSCO Future M, its battery materials subsidiary) has made Australian lithium investments that provide some upstream coverage for Korean producers, but the scale remains below CATL's integrated position.

Cathode Materials: Korea's Strategic Midstream

Korea has developed significant competitive advantage in cathode active materials (CAM) — the midstream processing step between raw minerals and battery cells. POSCO Future M, Ecopro BM, and L&F (now Lotte Chemical Battery Materials) are among the largest CAM producers globally, supplying high-nickel NMC precursor and cathode materials to Korean and international battery manufacturers. This midstream position creates some value capture even if upstream minerals are imported, but the cost structure still disadvantages Korea versus Chinese vertically integrated producers.

Part VII — Strategic Outlook: The Decade Ahead

2026-2028: Survive the Trough

The immediate period is a test of financial endurance. EV adoption growth in Europe and North America has disappointed versus forecasts, excess battery capacity has depressed spot prices, and Chinese price aggression continues. LG Energy Solution's IRA AMPC buffer provides the strongest near-term protection. Samsung SDI must manage its loss position while the ASSB programme matures. SK On must reach operational profitability consistently before SK Group's patience expires.

2028-2030: The ASSB Inflection

If any producer achieves commercial-scale all-solid-state battery production in this window — Samsung SDI is the most likely candidate among Korean producers — the competitive hierarchy shifts radically. ASSB-equipped vehicles would have 700-800km range, 10-minute fast charging, and zero thermal-runaway risk. OEMs that locked in ASSB supply contracts early would have a product advantage that justifies significant premium pricing. The winner of the ASSB race wins not just market share but pricing power — the single attribute most lacking in today's commoditising battery market.

The CATL Ceiling

CATL's dominance is real but not necessarily permanent. The company's primary advantages — scale manufacturing economics and Chinese domestic market size — are structural. But CATL's technology edge in advanced battery chemistry is narrowing. Korean HNM-based batteries still outperform CATL's NMC in cycle life at high charge rates. And CATL's access to the IRA-eligible US market is structurally constrained by FEOC provisions. In Europe, CATL's 2025 opening of its Hungary factory (CATL's first European plant) creates a direct Western manufacturing presence, but at higher cost than Chinese production.

The Korean battery empire will survive. Whether it thrives depends on the ASSB technology race, the continuation of IRA-equivalent policies under successive US administrations, and the financial endurance of SK On in particular to complete its investment cycle before market recovery.

Blue Lotus Research Intelligence | Northeast Asia Series | Korea Battery Empire | August 2026

All data sourced from public disclosures, SNE Research, BloombergNEF, and cited news sources.